



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO M/S HARAN FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

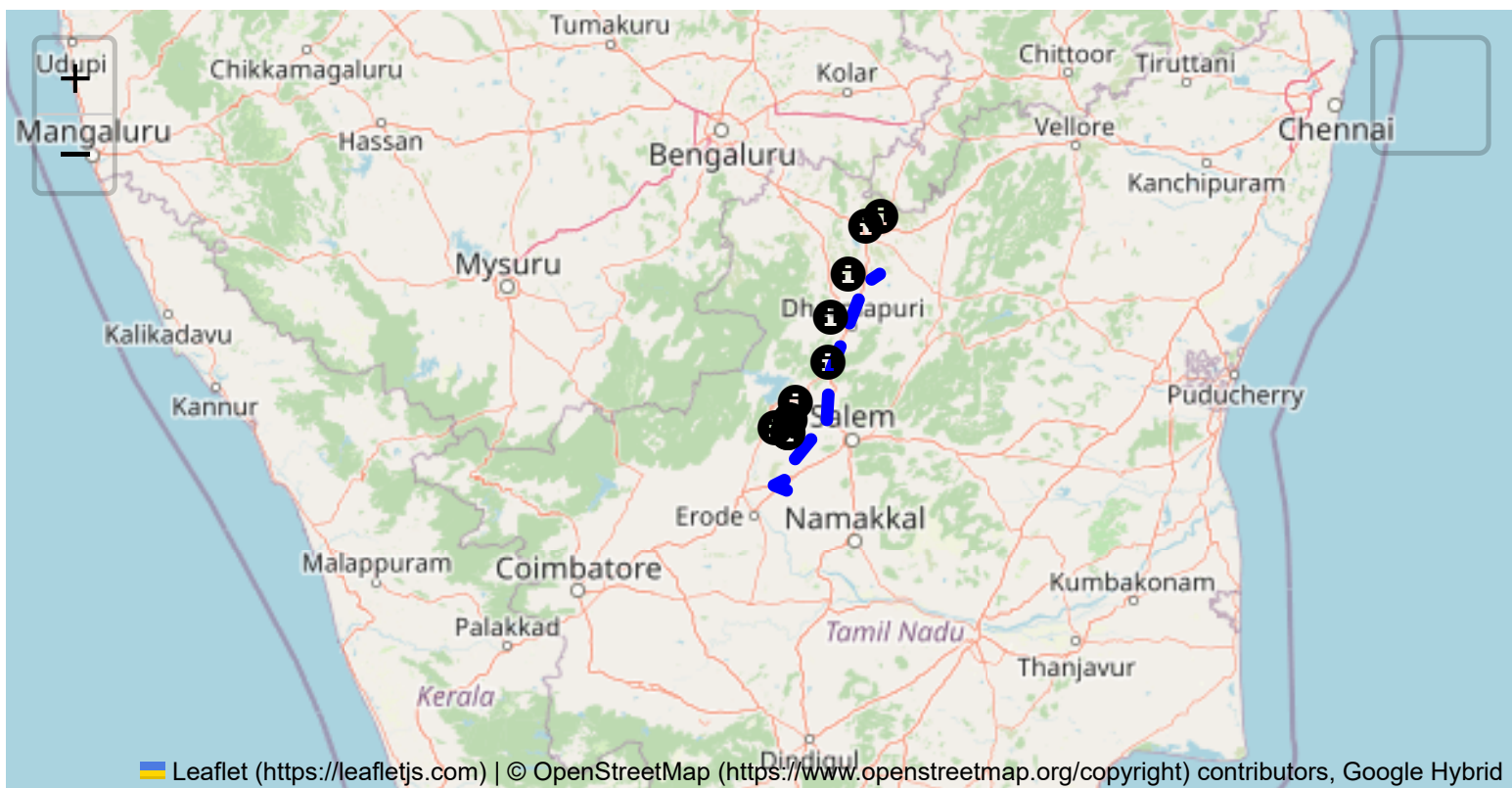
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

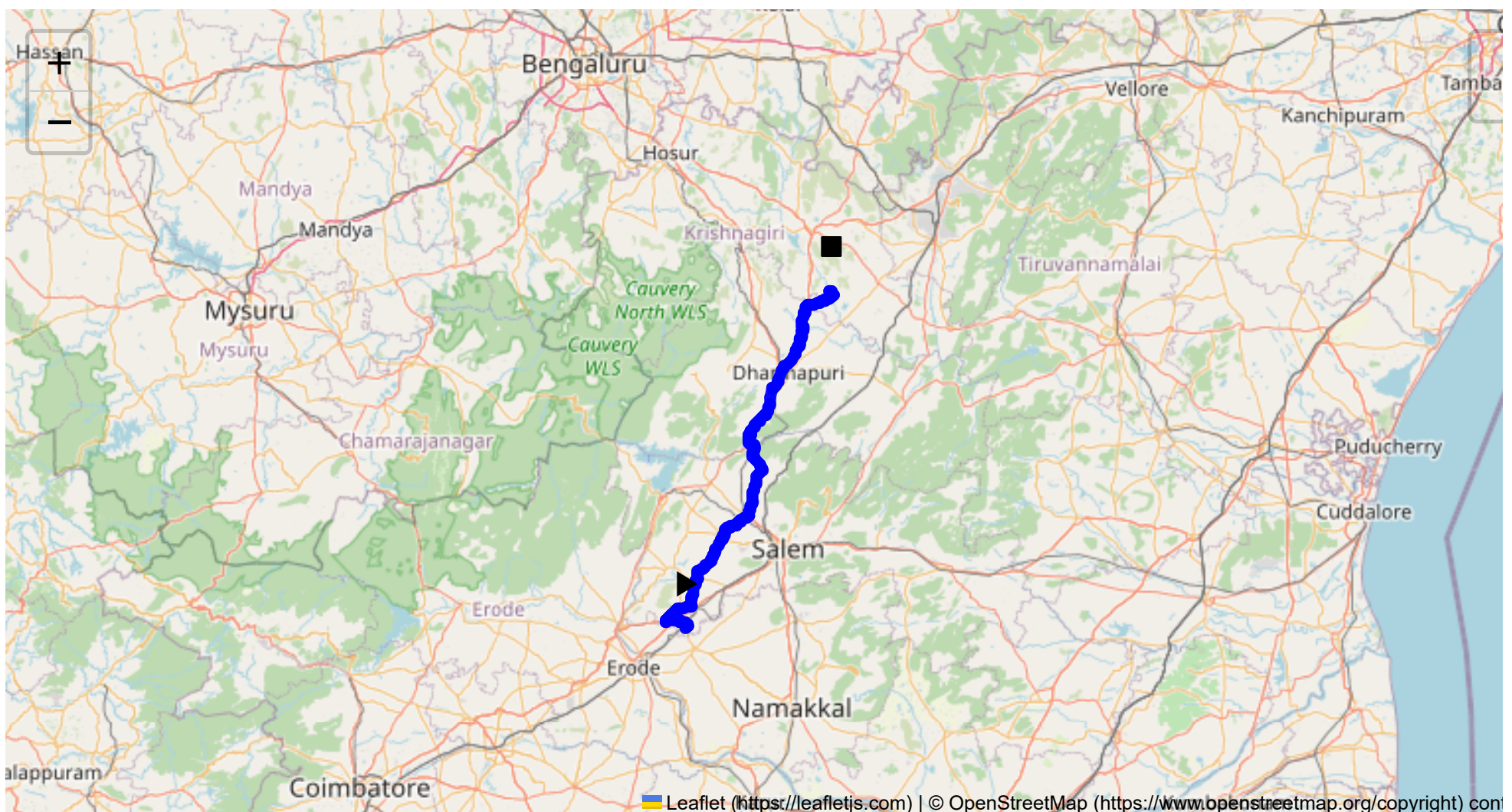
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 137.87 km
Estimated Duration: 2.8 hours
Adjusted Duration (Heavy Vehicle): 3.5 hours
Start: (11.4381, 77.8734)
End: (12.357483, 78.27479)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map:

- The route stretches approximately 137.87 kilometers and is traversed within Tamil Nadu. It connects Sangagiri to Maruderi via several key waypoints that include main highways and state roads. These roads are typically used for transporting goods between industrial and rural regions.

2. Typical Weather Conditions and Potential Weather-related Hazards:

- The region generally experiences hot and dry weather, especially from March to June. Monsoon season runs from October to December and can cause heavy rainfall, leading to potential flooding and slippery road conditions. Fog is a rare but possible occurrence in the winter months.

3. Analysis of Traffic Patterns:

- Traffic congestion is most prevalent during peak hours, typically from 8 AM to 10 AM and 5 PM to 8 PM, especially near major highways and city intersections. Areas like Salem and Konganapuram may witness heavier traffic due to local markets and junctions.

4. Assessment of Road Quality and Infrastructure:

- Overall road quality is moderate to good, with certain stretches, especially near urban centers, being well-maintained. However, rural and less frequented sections may have potholes and narrower lanes. Regular road maintenance checks are advised for accurate updates.

5. Suggestions for Alternative Routes for Emergencies:

- In emergencies, a viable detour involves using NH44, which runs parallel and is generally well-monitored. Local routes may provide alternatives, but should be verified for current conditions and accessibility.

6. Summary of Local Regulations Affecting Hazardous Material Transport:

- Transport of hazardous materials is regulated by national and state laws requiring permits for specific cargo. Night travel restrictions may apply, and adherence to speed limits and signage is crucial.

7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials:

- Historical data indicates occasional incidents involving vehicle overturns or leaks due to sudden maneuvers or mechanical failure. Awareness of hazardous material protocols is critical.

8. Environmental Considerations and Sensitive Areas:

- The route passes near agricultural zones and small water bodies. Care should be taken to avoid contamination, particularly with hazardous materials. Adequate spill management systems should be in place.

9. Analysis of Communication Coverage:

- Communication coverage is generally reliable along major highways. However, rural stretches may experience occasional dead zones. Installing additional signal boosters in vehicles may mitigate this issue.

10. Estimated Emergency Response Times for Different Route Segments:

- Emergency services are more readily available in urban areas like Salem, with response times ranging from 15 to 30 minutes. In rural areas, response times could extend to 45 minutes to an hour.

11. Overall Summary of Risk Assessment:

- The route is moderately safe with potential risks posed by weather conditions, rural road quality, and peak traffic congestion. Preparations for communication issues and knowledge of local regulations are vital for safe transport. Regular updates and road maintenance checks are recommended to ensure up-to-date safety measures.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43961, 77.87341	15 KM/Hr
1	Turn	High	11.44024, 77.87528	15 KM/Hr
2	Turn	High	11.44898, 77.87410	15 KM/Hr
3	Turn	High	11.45352, 77.85698	15 KM/Hr
4	Turn	Medium	11.45764, 77.85219	30 KM/Hr
5	Turn	High	11.46303, 77.84945	15 KM/Hr
6	Turn	Medium	11.45543, 77.81740	30 KM/Hr
7	Turn	High	11.45584, 77.81695	15 KM/Hr
8	Turn	Medium	11.45656, 77.81709	30 KM/Hr
9	Turn	High	11.49383, 77.88442	15 KM/Hr
10	Turn	Medium	11.56083, 77.89921	30 KM/Hr
11	Turn	Medium	11.56180, 77.90019	30 KM/Hr
12	Turn	Medium	11.59189, 77.90645	30 KM/Hr
13	Turn	Medium	11.59328, 77.90809	30 KM/Hr
14	Turn	Medium	11.63867, 77.94841	30 KM/Hr
15	Turn	High	11.64217, 77.95200	15 KM/Hr
16	Turn	Medium	11.64712, 77.95118	30 KM/Hr
17	Turn	Medium	11.67858, 77.96834	30 KM/Hr
18	Turn	Medium	11.70062, 77.97961	30 KM/Hr
19	Turn	High	11.70218, 77.97866	15 KM/Hr
20	Turn	Medium	11.71399, 77.99580	30 KM/Hr
21	Turn	Medium	11.71917, 78.01123	30 KM/Hr
22	Turn	Medium	11.73769, 78.03451	30 KM/Hr
23	Turn	Medium	11.73785, 78.03551	30 KM/Hr
24	Turn	High	11.73651, 78.03871	15 KM/Hr
25	Turn	Medium	11.74143, 78.04613	30 KM/Hr
26	Turn	Medium	11.74265, 78.04710	30 KM/Hr
27	Turn	Medium	11.78005, 78.05788	30 KM/Hr
28	Turn	Medium	11.86928, 78.08384	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
29	Turn	Medium	11.88009, 78.08164	30 KM/Hr
30	Turn	Medium	11.91277, 78.05853	30 KM/Hr
31	Turn	Medium	11.93746, 78.05331	30 KM/Hr
32	Turn	Medium	11.95531, 78.04698	30 KM/Hr
33	Turn	High	11.95854, 78.04990	15 KM/Hr
34	Turn	High	11.96015, 78.04945	15 KM/Hr
35	Turn	Medium	11.96370, 78.05142	30 KM/Hr
36	Turn	Medium	11.96433, 78.05113	30 KM/Hr
37	Turn	High	11.96542, 78.04938	15 KM/Hr
38	Turn	High	11.96650, 78.04927	15 KM/Hr
39	Turn	Medium	12.17027, 78.16812	30 KM/Hr
40	Turn	Medium	12.22496, 78.19149	30 KM/Hr
41	Turn	Medium	12.24670, 78.19551	30 KM/Hr
42	Turn	Medium	12.25108, 78.20005	30 KM/Hr
43	Turn	Medium	12.25411, 78.20024	30 KM/Hr
44	Turn	High	12.31467, 78.20768	15 KM/Hr
45	Turn	Medium	12.31791, 78.21461	30 KM/Hr
46	Turn	High	12.31908, 78.21586	15 KM/Hr
47	Turn	High	12.32377, 78.24215	15 KM/Hr
48	Turn	Medium	12.32526, 78.24360	30 KM/Hr
49	Turn	Medium	12.32942, 78.25439	30 KM/Hr
50	Turn	Medium	12.33765, 78.27106	30 KM/Hr
51	Turn	Medium	12.34288, 78.28170	30 KM/Hr
52	Turn	Medium	12.34494, 78.28231	30 KM/Hr
53	Turn	Medium	12.34543, 78.28209	30 KM/Hr
54	Turn	Medium	12.34597, 78.28120	30 KM/Hr
55	Turn	Medium	12.35207, 78.27707	30 KM/Hr
56	Turn	Medium	12.35220, 78.27673	30 KM/Hr
57	Turn	Medium	12.35636, 78.27477	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Sri Shanthi Hospital	11.569272, 77.901937	30 km/h	Medium
2	hospital	Omalur Government Hospital	11.74161, 78.044296	30 km/h	Medium
3	clinic	R.R. Nursing Home	11.7394129, 78.0439861	30 km/h	Medium
4	hospital	Sri Hari Hospital	11.7396726, 78.0440701	30 km/h	Medium
5	hospital	Sugumar Hospital	11.7396031, 78.0426955	30 km/h	Medium
6	hospital	Sri Sugam Hospital	11.7429821, 78.0461039	30 km/h	Medium
7	hospital	Saravana Hospital	11.7409303, 78.0458308	30 km/h	Medium
8	hospital	Arun Hospital	11.7429892, 78.0466881	30 km/h	Medium
9	clinic	Saratha Clinic	11.7422818, 78.0458175	30 km/h	Medium
11	clinic	Sri Meenakshi Nursing Home, Salem	11.7449162, 78.0474637	30 km/h	Medium
12	clinic	Divyam Child Care Clinic	11.7447135, 78.0473858	30 km/h	Medium
13	clinic	Momota Clinic	11.744982, 78.047397	30 km/h	Medium
14	clinic	Gomathi Clinic	11.7445293, 78.0470401	30 km/h	Medium
15	hospital	Vinayaka Child Hospital	11.745812, 78.0480225	30 km/h	Medium
16	hospital	Dr. G.R.Thilakam Hospital	11.746097, 78.04769	30 km/h	Medium
17	hospital	Thulasi Hospital	11.7461516, 78.0469984	30 km/h	Medium
18	hospital	Amirthalingam Hospital	11.7456708, 78.0470548	30 km/h	Medium
19	hospital	Children Hospital	11.7458377, 78.0480205	30 km/h	Medium
20	clinic	Shri Ramana Poly Clinic	11.7477122, 78.0479293	30 km/h	Medium
21	hospital	Dr. Nagaraj Hospital	11.7464035, 78.0472401	30 km/h	Medium
22	clinic	Manimuthu Skin Clinic	11.7521189, 78.0484058	30 km/h	Medium
23	hospital	St. Joseph's Speciality Hospital	11.761611, 78.0522344	30 km/h	Medium
25	hospital	Anugraha Multi Speciality Hospital	11.773549, 78.0546408	30 km/h	Medium
27	hospital	L.R. Hospital	11.8703354, 78.0844554	30 km/h	Medium
28	clinic	JSA Clinic	11.8700619, 78.0844982	30 km/h	Medium
29	hospital	Thoppur,Government Hospital	11.9398824, 78.0551658	30 km/h	Medium
30	hospital	Government Hospital	12.3048002, 78.2060881	30 km/h	Medium
32	clinic	G.K Clinic - Beruhalli	12.3566347, 78.2746656	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
10	school	Fathima Hr Sec School	11.7410508, 78.0479272	30 km/h	Medium
24	school	John Britto Matriculation Higher Secondary School	11.7676907, 78.0513092	30 km/h	Medium
26	college	Dhirajlal Gandhi College of Technology	11.7968014, 78.0562095	30 km/h	Medium
31	school	Government Boys Higher secondary school	12.3047059, 78.2065495	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44024, 77.87528



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45764, 77.85219



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45543, 77.81740



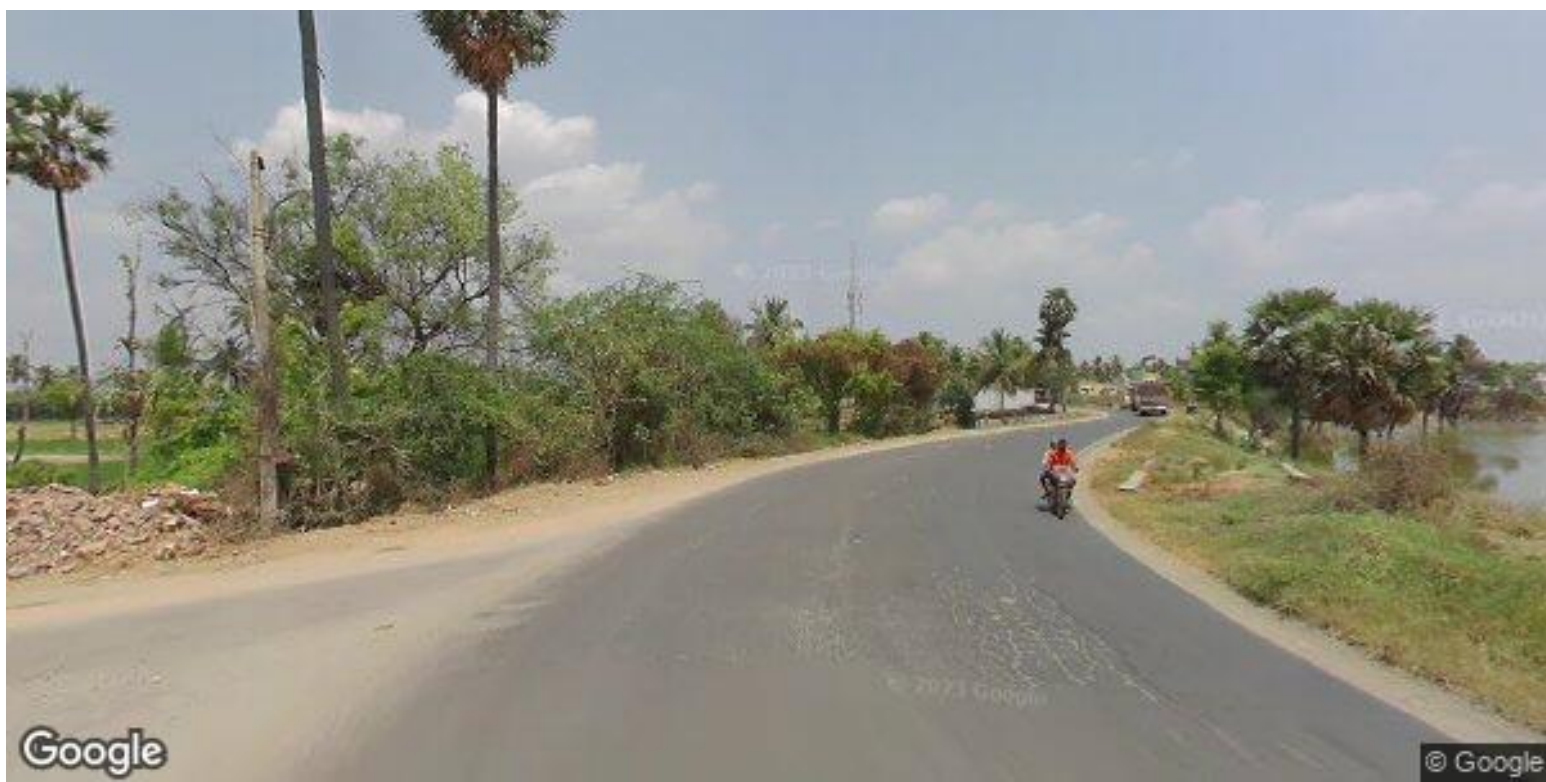
Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45584, 77.81695



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45656, 77.81709



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49383, 77.88442



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56083, 77.89921



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.56180, 77.90019



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59189, 77.90645



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.59328, 77.90809



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.63867, 77.94841



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.64217, 77.95200



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64712, 77.95118



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67858, 77.96834



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.70062, 77.97961



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70218, 77.97866



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71399, 77.99580



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.71917, 78.01123



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73769, 78.03451



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73785, 78.03551



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.73651, 78.03871



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74143, 78.04613



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74265, 78.04710



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.78005, 78.05788



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.86928, 78.08384



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.88009, 78.08164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.91277, 78.05853



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.93746, 78.05331



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.95531, 78.04698



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.95854, 78.04990



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.96015, 78.04945

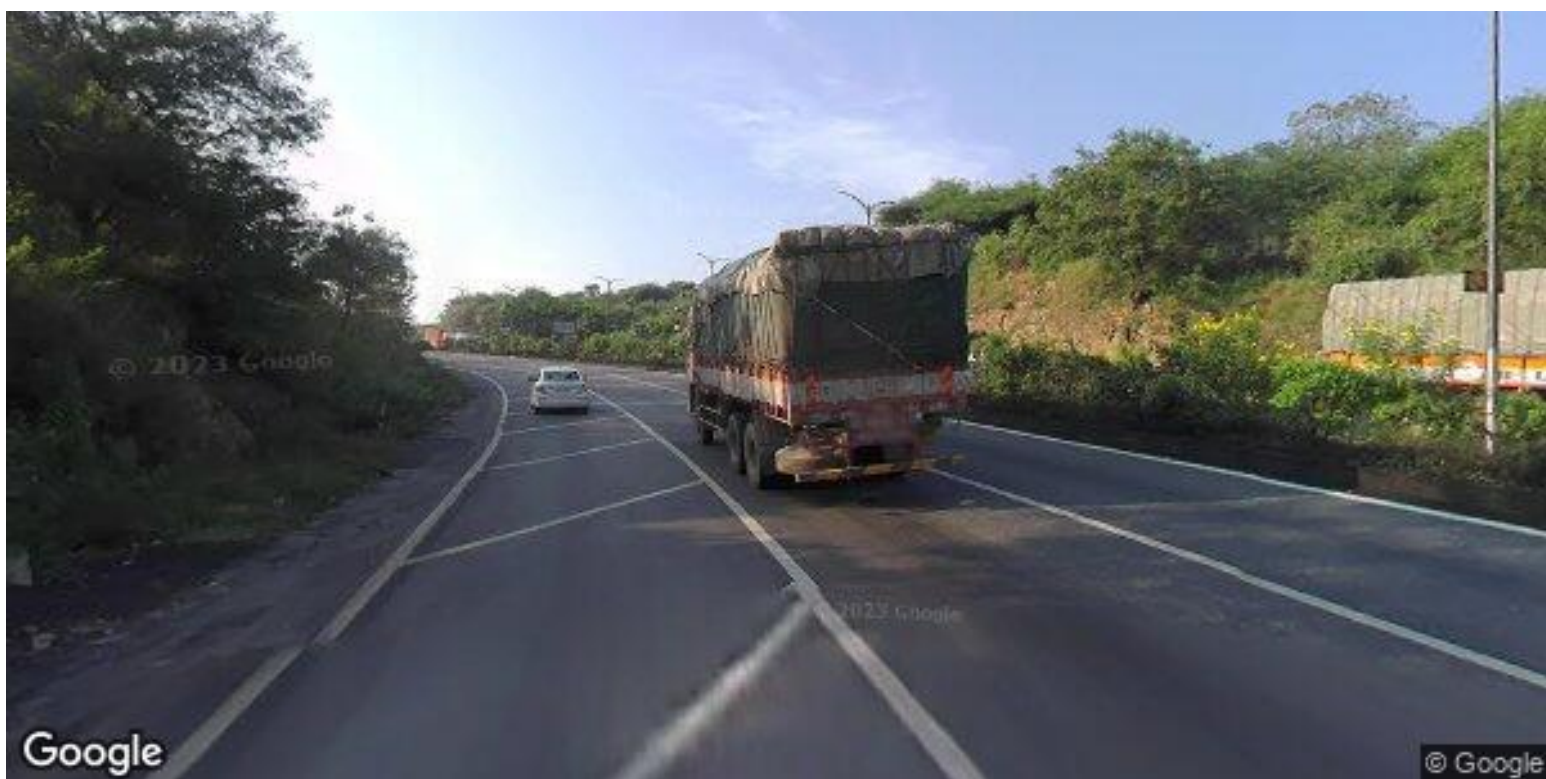


Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96370, 78.05142



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.96433, 78.05113



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96542, 78.04938



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.96650, 78.04927



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.17027, 78.16812



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.22496, 78.19149



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.24670, 78.19551



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25108, 78.20005



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 12.25411, 78.20024



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 12.31467, 78.20768



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.31791, 78.21461



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.31908, 78.21586



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 12.32377, 78.24215



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.35207, 78.27707



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.35220, 78.27673



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 12.35636, 78.27477
